International Journal of Hospitality Management 124 (2025) 103987



Contents lists available at ScienceDirect

International Journal of Hospitality Management

journal homepage: www.elsevier.com/locate/ijhm



A study of employee attitudes towards AI, its effect on sustainable development goals and non-financial performance in independent hotels

Maria Jesus Jerez-Jerez

University of Bedfordshire, Vicarage St, Luton LU1 3JU, United Kingdom

ARTICLE INFO

Keywords:
Sustainable development goals
Artificial intelligence
Employee readiness
Employee acceptance
Non-financial performance, independent hotels

ABSTRACT

This study explores the effect of hotel employees' readiness for and acceptance of Artificial Intelligence (AI), on hotels' adoption of AI, and its subsequent impact on achieving Sustainable Development Goals (SDGs), as well as impact on non-financial performance (NFP), within the U.S. independent hotel sector. A novel survey instrument was devised, validated and administered to 1600 employees in independent hotels across the United States. Structural Equation Modelling (SEM) was employed to test the hypotheses derived from a conceptual framework. The results confirmed that employee readiness and acceptance of AI significantly affects AI and SDG adoption, and also positively impacted NFP metrics such as employee optimism, satisfaction and engagement. The study finds evidence of a pathway from employee engagement with AI to greater SDG adoption, and in turn, enhanced NFP. This highlights the significance of leveraging employee attitudes toward AI for more sustainable and effective performance in the hospitality sector.

1. Introduction

The rapid proliferation and use of artificial intelligence (AI) e.g. Chatbots, Virtual Assistants, Automated Hotel Management Systems, predictive analytics, etc. is now becoming a reality across many aspects of business and society in general. AI refers to the simulation of human intelligence processes by computer systems. The use of AI in the workplace can help to automate tasks, permitting time for employees to focus on more complex or creative work. AI can also support better decision-making by allowing employees to analyse more data and draw more accurate conclusions (Tschang and Almirall, 2021). To that end, hoteliers have begun adapting and adopting AI not only to deliver a more efficient customer experience, but also as part of employee experience management (the strategic management of employees' internalisation and interaction with their organisation, improving their experience and output), with many incorporating the notion into their business mission statements. Equally, hotels around the globe are aware of the new possibilities that AI has to offer (e.g. Voice Activated Assistants, AI remote Check-in, and Property Management Systems (PMS), facilitating workforce management, forecasting staffing requirements and supporting training and development for employees) and have started exploring them (Buhalis et al., 2019).

One of the key drivers of hotels' success is the ability to tackle an increasing customer demand for a seamless and personalised experience from booking, arrival and throughout the duration of their stay (Barthel et al., 2015). This has amplified the pressures on hotel employees to maintain flawless, bespoke experiences for customers. Nevertheless, these customer-oriented demands are not the only forces behind an increased use of AI. UN mandated SDGs have also further accelerated AI adoption and use (Rydzik and Kissoon, 2021) to support wide-ranging health, safety, environmental and community initiatives emerging in several sectors. The expansion of sustainably directed operations across all levels of business subsequently impacts employees, who are also increasingly aware of the need for improved sustainability. Therefore, employee readiness and acceptance of AI could be major contributing factors to its adoption which, in turn, may have a determining impact on hotels' achievement of SDGs and non-financial performance. According to a report by McKinsey, AI disruptive technologies are forecast to have a positive $6.2 trillion effect on the world economy by 2025, with hospitality as one of the key industries to be impacted (Manyika et al., 2013). However, this presents both challenges and opportunities for hoteliers. Therefore, it is important that hotels assess the real value that using AI can have on both SDG adoption and NFP. This relies on an understanding of their employees' working experience in the hotel, and of their acceptance of and emotional response to AI (Hwang et al., 2021) to best inform management strategy.

Despite the emergence of AI and the recognition by practitioners of its potential importance in influencing SDG adoption and ultimately

E-mail address: Maria.jerez@blueyonder.co.uk.

https://doi.org/10.1016/j.ijhm.2024.103987
Received 15 May 2024; Received in revised form 12 August 2024; Accepted 23 October 2024
Available online 28 October 2024
0278-4319/Crown Copyright © 2024 Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

NFP, academic literature investigating this topic has been limited. Most studies about technology adoption have looked through the consumer lens only so far (e.g. Skandali et al., 2024), while publications on the management of employee experience and impact on NFP are mainly found in practitioner-oriented journals or management books (e.g. Smerecnik & Andersen, 2011). Furthermore, extant literature (e.g. Hwang et al., 2021) primarily focuses more on managerial actions and outcomes, rather than on the theories underlying the antecedents and consequences of employee experience. To date, rigorous empirical studies investigating how employee experience and the inherent value of employee readiness, acceptance, and response to AI adoption as a means of promoting SDGs, are lacking, leaving a significant gap in literature. However, a number of employee-focused studies (e.g. Di Vaio et al., 2022) have considered the role of non-financial reporting in fostering the achievement of SDGs, or the importance of exploring AI adoption through both linear and non-linear relationships, highlighting the influencing role of perceptual and attitudinal factors in this process. These form a basis for our further exploration, providing relevant starting points to improving the knowledge in this field. Furthermore, although several authors have studied SDGs within multinational hospitality companies (Rydzik and Kissoon, 2021), there is limited research focusing on independent hotels. To address this gap, the focus of this research is on employee perceptions regarding SDGs in independent hotels in the US. In doing so, it provides insight for hoteliers and managers into the value created by using AI in the hotel environment and how this relates to the employee experience.

The purpose of this study is therefore to investigate the effect of hotel employee readiness for and acceptance of AI, on AI and SDG adoption and NFP. Drawing on the literature focusing on employee experience, this research proposes and tests the notion that AI adoption with a view to improving SDG commitment (e.g. health and safety, and environmental initiatives) may contribute to a positive impact on NFP. By drawing on multiple technology adoption theoretical frameworks, this study aims to provide a more comprehensive understanding of AI adoption in the hospitality industry, offering new insights into both the antecedents and consequences of employee engagement with AI. Also, through its focus on independent hotels, it has the potential to make unique contributions that have been previously overlooked by the literature. Two primary research questions are addressed, with the goal of aiding hotel managers in understanding how hotels' SDG adoption and NFP are influenced by the acceptance and use of AI among employees: (1) To what extent does employee readiness for and acceptance of AI affect AI adoption? (2) How are hotels' SDG adoption and NFP affected by the adoption of AI? A novel SDG and AI adoption scale was developed and validated to address these questions.

2. Literature review

2.1. AI adoption and SDG commitment in hospitality

Studies show that SDG-related company activities have positive impacts on employee behaviours (de Visser-Amundson, 2022). Moreover, SDG adoption often revolves around economic, environmental, and social values (e.g.: Oriade et al., 2021). Organisations that actively promote and align with such values encourage a stronger sense of purpose and identity among employees (Salameh et al., 2023). This sense of contributing to a larger societal goal by virtue of active commitment to SDGs can be reinforced by use of AI and may therefore also boost employee optimism, engagement (Guix and Font, 2020), and NFP since improved engagement tends to improve employee performance in all areas of work. Hotel managers would benefit from a better understanding of the facilitating role that employee experience management, particularly in relation to readiness and acceptance of AI for the purposes of improved SDG commitment, can play in enhancing NFP.

As consumers are increasingly concerned with sustainability (Sun et al., 2021), more firms are paying attention to SDGs, causing the global sustainability movement to gain momentum (e.g.: Gonzalez-Arcos et al., 2021). Not surprisingly, achieving significant contribution to the SDGs specified by the UN can be challenging. First, the extensive scope and complexity of the 17 SDGs, along with 232 indicators and 169 targets, can easily become overwhelming for firms (Nilsson et al., 2018). Secondly, as SDGs are designed as national targets (Montiel et al., 2021), it may be hard for firms to interpret and integrate them into their business strategies. International business researchers could benefit immensely from the development of a measurement instrument to identify strategies that contribute to achieving SDGs (Cuervo-Cazurra et al., 2021). In hospitality, research suggests that adoption of SDGs by organisations within their operations has become imperative to enhance competitiveness, improve stakeholder relations and contribute to the business' and even the sector's long-term sustainability. Although SDG adoption is more widespread among the larger hospitality chains, significant challenges remain for smaller and independent organisations due to the scope, diversity, and complexity of achieving the SDGs, with considerable difficulty in translating the broad SDG targets into actionable strategies for these smaller businesses (Legrand and Matthew-Bolofinde, 2022).

2.2. The role of AI and SDGs in enhancing employee engagement and NFP

Marketing and chief executives' agendas have recognised the importance of understanding and enhancing NFP, in hotels and hospitality broadly (Lim et al., 2023), and this remains a critical field for academic studies. As highlighted by Pantano and Timmermans (2014), the relationship between employees and hoteliers is strengthened by employees' emotional engagement with the organisation. This can be supported by easily useable interactive technology and interfaces, helping employees perform everyday roles and duties, which also allow employees and hoteliers a dynamic, reflexive and communicative exchange and visualisation of information. So, the use of AI has the potential to connect hoteliers and employees, improving NFP (Buhalis et al., 2023). For example, predictive analytics permit data analysis and the forecasting of consumer behaviour and market trends, informing data-driven decisions, effective marketing strategies and the automatisation of tasks, leading to more efficient working environments and more opportunities for hospitality employees to focus on the direct connection with customers. Moreover, AI-powered operational decisions for the purposes of improving sustainability can reduce duplication and waste, as well as developing communication with employees, producing a greater sense of engagement with the organisation and its goals. Supporting this, employee AI readiness is fundamental in this engagement, with three main three elements: 'employee self-efficacy' which reflects confidence in using AI, 'role clarity' defines understanding of how AI integrates into employee responsibilities, and 'motivation' represents the employee's drive around AI adoption. This research considers these elements, to study the relationship between employee perception of (and willingness to adopt) emerging AI-based strategies, and improved NFP and SDG adoption, in a hotel environment.

Arguments with respect to the importance of NFP and its relationship to AI are widely supported in the literature discussing employee acceptance of AI in the hotel environment, emphasising the increasing significance of employee engagement with the policies, operations and goals of their hotel, and its impact on NFP in a fast-changing digital environment (Anderson and Bolton., 2015). Studies (e.g.: Molina--Collado et al., 2022) support the view that organisational focus on the improvement of employee engagement is a critical facet of SDG adoption, supplemented and supported by AI, stressing the ability of hoteliers to get to grips with developing technologies. Moreover, the ever-expanding role of AI in all aspects of business may be daunting for hoteliers unless they are able to embrace technological change and support an employee experience that capitalises on the use of AI to improve overall employee engagement with the organisation and its priorities. In so doing, employees are provided a greater sense of

involvement in SDG adoption and NFP improvement goals, reinforcing commitment at all levels of the organisation.

There is considerable scholarly literature on NFP, and researchers of marketing have explored how the transition to digitalisation is impacting NFP (e.g.: Cakir et al., 2019). More recently, as the number of contact points between a business and its customers has increased, especially with the rise of AI, research has revealed the essential importance of monitoring the many employee-consumer interaction experiences that are created from those contact points, as these play a significant role in influencing the consumers' brand preferences, which then impact on consumers' purchase decisions, and ultimately customer retention (Kim et al., 2019). But more importantly, these contact points also offer improved opportunity for management to develop a nuanced understanding of their employees' NFP, using this data to drive improved employee management and communication, increasing employee engagement. A study by Anderson and Bolton (2015) highlighted the importance of the use of AI such as sensors and radio frequency identification (RFID) within the hotel sector, to capture data to be interpreted for retail acumen. The sensors capture simple data, such as the number of customers who have walked through a doorway or down an aisle, to more complex data, such as demographic or behavioural data. For a hotelier, this provides an opportunity for rich data analysis and insights that can be communicated to employees, enhancing employee management and directly driving NFP as well as sales (Anderson and Bolton., 2015). For instance, hotel management can analyse the data provided by AI to inform employees of the appropriate kind of customer engagement and sales methods best suited to specific situations, or simply direct staff to focus on the areas of highest or lowest customer traffic based on set targets, thus increasing their work efficiency, service quality and henceforth work satisfaction (Nam et al., 2020).

According to Pantano and Timmermans (2014), the implementation of AI in hotels necessitates modifications in both selling activities and businesses processes. The authors highlight that from an organisational point of view, recognising, selecting and employing the optimal AI technology requires time and effort, but can provide an effective means of obtaining, managing and transferring knowledge from customers to companies and vice versa. As a result, studying employee readiness for AI adoption, alongside leveraging the most appropriate AI, has become essential. Scholarly studies such as Foroudi et al. (2018) demonstrate that education and innovation efforts from which a company may gain advantage, need not essentially be located within the business and may well be based within the customer environment. Consequently, by using the most appropriate AI, a smart partnership between hotelier, employee, and customer is created. For instance, a simple albeit artificially intelligent automation of repetitive tasks can allow hotel employees to focus on more strategic concerns including developing their relationships with the consumer. However, there is a need for hoteliers to understand employee demands and their acceptance of AI (Chang et al., 2016) such as how employees perceive the usefulness and ease of use of AI, which are necessary prerequisites for the adoption of AI in the context of SDG.

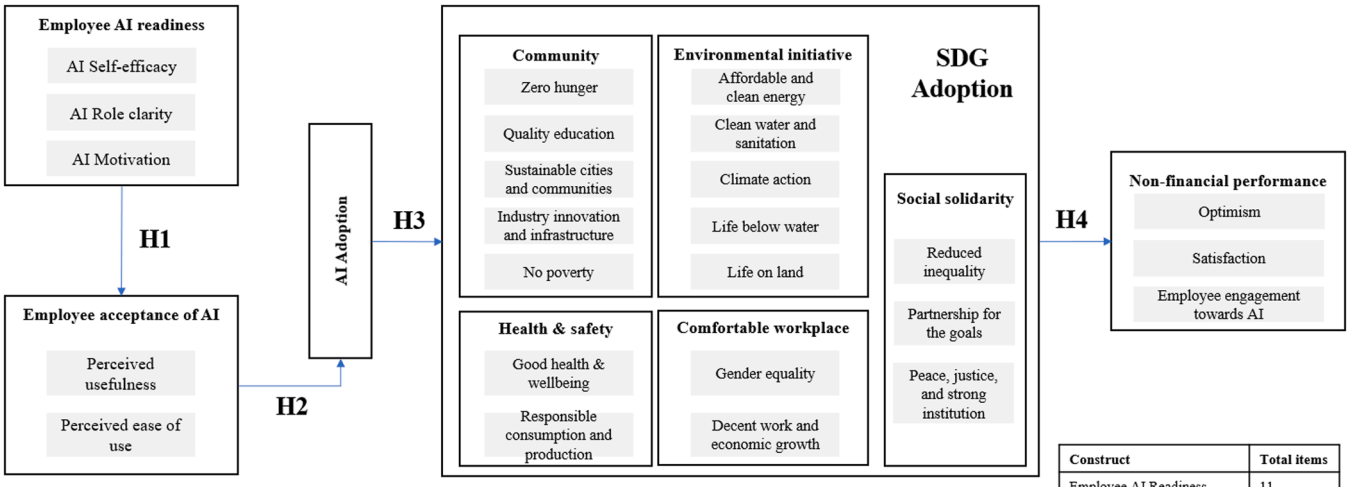
3. Explanation of the conceptual framework

The conceptual framework (see Fig. 1) is derived from an interlinked set of constructs from the prior literature and theoretical models used by other scholars in similar studies (e.g. TOE – Technology-organisation-environment - Chititipaka et al., 2023 and TAM – Technology Acceptance Model - Kamal et al., 2020). The level of employees' readiness and acceptance toward AI (including self-efficacy, roles clarity and motivation) influence the AI adoption and its effect on the adoption of the SDGs and non-financial performance (Foroudi et al., 2023). Employee self-efficacy reflects confidence in using AI, role clarity defines understanding of how AI integrates into their responsibilities, and motivation represents the drive towards AI adoption. These factors, along with the perceived usefulness and ease of use of AI, form the basis for employee acceptance, essential for effective AI adoption. This study then hypothesises that AI adoption is a key precursor for the implementation of sustainable practices aligned with the United Nations SDGs (in our conceptual framework, we consider Community, Health and Safety, Environmental Initiative, Comfortable workplace, and Social solidarity-as these are the SDGs most relevant to the hospitality sector -see Fig. 1), which is then hypothesised to have a positive impact on NFP measures including employee optimism and satisfaction, as well as (in turn) employee engagement with AI, thereby addressing significant gaps identified in the literature, using experimental methods.

3.1. Hypotheses

3.1.1. Employee readiness → Employee acceptance of AI

The value of employing AI devices lies in their ability to enhance



Construct	Total items
Employee AI Readiness	11
Employee Acceptance of AI	9
AI Adoption	5
SDG Adoption	65
Non-financial performance	7

Fig. 1. Conceptual framework.

both efficiency and effectiveness, as observed in hospitality settings (Ribeiro et al., 2022). Employees exhibit various behavioural responses (avoidance, approach, and neutral) toward novel or high-tech products and services (Wang et al., 2017), informing favourable or unfavourable views about new technology-based products and services. Such views are dependent on the level of employee AI readiness and here we will consider its drivers, including characteristics such as employees' level of self-efficacy, understanding of job role, and motivation to engage with AI. The level of preparation and willingness of employees to embrace AI is based on their subjective response and perceived assessment of the AI features (Yang et al., 2015), and it can be influenced by elements such as individuals' fear of change, ethical concerns regarding AI and their understanding of the technology and how to interact with it (Islam et al., 2021). In hospitality specifically, the need for face-to-face human interaction as the essence of customer service is another aspect to acknowledge, potentially hindering employees' acceptance of AI since the implementation of novel technology might necessarily reduce this interaction (Sun et al., 2023). AI readiness is regarded as a main driver of employee acceptance of AI based technologies (Kim et al., 2019). Employees showing different levels of AI readiness tend to express different intent, and behaviour when exposed to AI devices in their work (Rather et al., 2023). Thus, it is expected that when employees show higher level of AI readiness, they are more willing to accept AI technologies. As such, we propose the following hypothesis:

H1. : Employee readiness (self-efficacy, role clarity, motivation) for AI positively impacts on employee acceptance of AI, in independent hotels

3.1.2. Employee acceptance of AI ➔ AI adoption

Factors underlying adoption of AI technologies have attracted attention from different scholars across different disciplines. From the use of generative AI in creative industries (Bhusan, 2024), to large-scale AI adoption in public administration, food manufacturing or healthcare sectors (e.g.: Wang et al, 2023), AI adoption has been found to be influenced by many factors across various domains, including organisational (e.g. level of company resources and support), technological (e.g. complexity and ease of use), environmental (e.g. social influence, ethics of use) and human (e.g. level of employee training). Due to the direct impact on guest interaction, employee attitudes about the adoption of AI technologies in hospitality-related services may vary more than in non-service industries.

Although some employees might anticipate improved service quality and efficiency due to the integration of advanced technologies (Zalama et al., 2014), they might also experience feelings of detachment and discomfort due to the human-like features of AI devices (Murphy, 2000). So, AI adoption hinges on employees perceiving clear benefits in their tasks, and in relation to guests' experiencing enhanced service, with the perceived ease of use being as crucial as AI's practical utility (Chatterjee et al., 2021). However, the literature is limited around how employee self-efficacy, role clarity, motivation, and perceptions of AI's usefulness and ease of use contribute to its acceptance and adoption within independent hotels. This study seeks to fill this gap, as such, we propose the following hypothesis:

H2. : Employee acceptance of AI (influenced by perceptions around ease of use and usefulness) positively impacts on AI adoption, in independent hotels

3.1.3. AI adoption ➔ SDG adoption

Hotels are drawn to digital technologies, as they are perceived as an effective means to achieve SDGs, and they can be integrated into products, services, procedures, and business frameworks (Stylos et al., 2021). Moreover, these AI technologies encompass capabilities such as experiential learning, independent environmental awareness, decision-making, and interaction with humans and other automated systems (Salehi et al., 2021). These innovations enhance SDGs by averting or reducing pollution, or efficiently using resources like energy (Jang et al., 2024). The global adoption of AI solutions has greatly influenced sustainable development by enabling increased sustainability, improved efficiency, and enhanced effectiveness in hospitality setting (Li et al., 2021). From a technical standpoint, developers and designers have actively created technologies that reduce their carbon footprint and carbon emissions and promote solar energy use (Buhalis and Leung, 2018). Additionally, concerted efforts have been made to develop applications that aid waste management, water supply and management, and crop production. In the hospitality and travel setting, AI-driven systems can optimise energy consumption by analysing occupancy patterns, weather forecasts, and guest preferences (Ribeiro et al., 2022). This can lead to more efficient heating, cooling, and lighting control, reducing energy wastage and environmental impact. Moreover, AI can aid hotels in connecting with local communities to support sustainable initiatives and create shared value (Du et al., 2022). As such, we propose the following hypothesis:

H3. : AI adoption positively impacts on SDG adoption, in independent hotels

3.1.4. SDG adoption ➔ Non-financial performance

Studies show that SDG-related firm activities have positive impacts on employee behaviours (de Visser-Amundson, 2022). These activities are often considered admirable (de Ruyter et al., 2022), and drawing on the foundation of moral psychology theory (Haidt, 2007), business activities and decisions that are perceived as morally praiseworthy may encourage employees to exhibit favourable behaviours towards an organisation. SDG adoption revolves around positive economic, environmental, and social values (e.g.: Oriade et al., 2021). When employees are part of an organisation that actively promotes and aligns with such values, they tend to develop a stronger sense of purpose, and identification with their work (Salameh et al., 2023). This sense of contributing to a larger societal goal can boost their optimism and engagement (Guix and Font, 2020). AI, however, has a specific application in the context of independent hotels, even though previous research has demonstrated the positive impact of SDG-related activities on employee perceptions and behaviour. We aim to offer new insight into how the SDGs can influence employee perceptions and NFP of independent hotels, highlighting their unique dynamics. As a result, we propose the following hypothesis.

H4. : SDG adoption positively impacts on the employee non-financial performance (optimism, satisfaction, employee engagement towards AI), in independent hotels

While the first two hypotheses relate to employees directly, the last two relate to the companies they work for. This approach allowed us to directly link employee perceptions and behaviours with outcomes at the organisational level, by aggregating individual responses to infer firm-level data.

4. Methodology

4.1. Quantitative research approach and scale development

We took a quantitative research approach, which necessitated the development of a novel SDG and AI adoption multi-items scale that evaluates employees' perceptions in the hospitality sector. Our scale was built based on existent measures from previous literature encompassing hospitality, sustainability, management, marketing, and organisational studies from various perspectives (e.g.: Hair et al., 2018). Measures assessing Employee AI readiness were adopted from Kohler et al. (2011) and Meuter et al. (2005), encompassing three dimensions: self-efficacy, role clarity and motivation. The Employee Acceptance of AI was measured based on Davis (1989), Venkatesh and Bala (2008), also encompassing two dimensions: perceived usefulness and perceived ease of use. AI Adoption was measured using scales adapted from Hopp and

Gangadharbatla (2016), while measures assessing SDG Adoption were adapted from Foroudi et al. (2023), encompassing 5 sub-constructs (community, health and safety, environmental initiative, comfortable workplace and social solidarity), each with their respective dimensions, reflecting the 17 UN SDGs, as presented in the conceptual framework (Fig. 1, which also details the number of items per construct). Lastly, NFP was measured using scales adapted from Zaichkowsky (1985) and Christ-Brendemühl, and Schaarschmidt (2020), encompassing three dimensions: optimism, satisfaction and employee engagement with AI (see Appendix B for more details on the constructs). The final questionnaire was well designed with a clear shape, layout, structure and visual appearance for the respondents to easily complete (Schaefer & Diffman, 1998). A seven-point Likert scale (Churchill & Peter, 1984) was used for all questions.

4.2. Data collection

The scale tested 5 key constructs in total: Employee AI readiness, Employee AI acceptance, AI adoption, SDG Adoption (including 5 sub-constructs drawn from the UN SDGs - Community; Health and Safety; Environmental Initiative; Comfortable Workplace and Social Solidarity - representing those relevant to the specific sustainability issues pertinent to the hospitality industry), and Non-Financial Performance, each with their relevant dimensions and items measured, as set out in Fig. 1. For the main study, 1600 anonymous questionnaires were used to collect data from the employees of independent hotels in the US. The surveys were administered during December 2023 to adults across the U.S. using the online platform Amazon Mturk (Chan and Wan, 2023). The following recruitment criteria was used: employees working in full time positions with AI interaction (e.g. front desk, customer service, housekeeping management, facilities management, etc.), part timers and non-operational roles being excluded from this research. Independent hotels were identified as the ideal unit of analysis given their unique and flexible ways of working, propensity of more limited resources, adaptability to change and increased need for customised solutions versus larger hotel chains (Nazarian et al., 2019). The US was chosen as research setting due to the large number of independent hotels: 40 % of hotels were independent in 2019 and as of 2021, there are 4600 boutique hotels in the US (Hotel Tech Report, 2024).

5. Data analysis and results

5.1. Pilot study

In our pilot study, the reliability of the scale was ensured, firstly through a face validity check with experts in the field of hospitality and sustainability. Furthermore, we refined the scale with data from 294 sample employees from independent hotels staff in the U.S., collected using online Amazon Mturk. This was used to refine the scale using statistical methods to confirm its validity (Denscombe, 2007). Three cases were removed due to missing data and low-quality responses. 275 questionnaires were therefore used to assess the reliability of the measures. Cronbach's alpha was calculated for each construct. The result showed Cronbach's alpha of 0.748 and higher. Four items were deleted (SZH1; SGH1; SGE4). According to scholars (Hair et al., 2018) Cronbach's alpha that is greater than 0.70 shows a high degree of reliability. The 275 questionnaires from the pilot data were not included in the final dataset.

5.2. Main study

5.2.1. Data preparation

The data was analysed in three steps: through Exploratory Factor Analysis (EFA), Confirmatory Factor Analysis (CFA) and finally Structural Equation Modelling (SEM). However, prior to these, the data was quality checked. We obtained 1543 complete and usable questionnaires for inferential analysis. Furthermore, we examined 50 early and 50 late participants and confirmed that there were no non-response bias or similar issues. In the final dataset (N=1543), 43.6 % identified as female and 55.8 % as male, 33.5 % fell within the age range of 24–30, 31.5 % between 31 and 39, while 28.4 % were aged between 40 and 59. Moreover, 28.3 % held postgraduate degrees and 65.7 % possessed undergraduate degrees, with 43.1 % occupying Middle Manager roles, while 20.4 % occupying Senior Manager roles, and having worked at the company for 3–5 years (35.7 %) or 5–10 years mostly (28.1 %). Full demographics can be found in Table 1.

5.2.2. Exploratory factor analysis (EFA)

To diminish the number of scale items, we conducted an Exploratory Factor Analysis (EFA). This step was required even after the pilot study since we had a larger sample size that allowed us to refine our measurement scales further. The number of extractable factors was determined by the latent root criterion (eigenvalue $>$ 1.00) (Hair et al., 2018). This analysis yielded seven solutions with eigenvalues surpassing 1, accounting for a total variance of 67 %, and the factors were deemed significant based on a scree plot. We carefully assessed items with loadings greater than.62 to interpret the solutions. These extracted factors reflected the constructs of the conceptual model closely hence making our scale congruent with the theory developed for this study. Additionally, we evaluated the internal reliability using Cronbach's alpha. Following the approach of other researchers (Brakus et al., 2009), items were excluded if they were not rated by at least 10 % of the participants, on the assumption that they might have been unclear. It is customary in scale development to allow for some missing data to prevent possible bias from ambiguous items and uphold the applicability and validity of the scale.

5.2.3. Confirmatory factor analysis (CFA)

In this phase the researchers used CFA to evaluate the 17 factors related to the Sustainable Development Goals (SDGs) which included a total of 77 items along, with the other four key constructs. The scale developed to measure AI adoption in this study demonstrated reliability indicated by both a Cronbach's alpha coefficient and Composite reliability (CR) exceeding the threshold of 0.70 suggesting that the constructs were reliable for research purposes (Hair et al., 2018). It is worth

Table 1 –
Demographics of the final study sample.

Category	Characteristics	Frequency	Relative Frequency (%)
Gender	Male	861	55.80
	Female	674	43.68
	Bisexual	4	0.26
	Lesbian	2	0.13
	Transsexual/Transgender	1	0.06
	Gay	1	0.06
Age	18–23	36	2.33
	24–30	518	33.57
	31–39	487	31.56
	40–59	439	28.45
	60-above	63	4.08
Education	Below Undergraduate	91	5.90
	Undergraduate	1014	65.72
	Postgraduate and above	438	28.39
Current Position	Professional (no management)	279	18.08
	Junior Manager	202	13.09
	Middle Manager	665	43.10
	Senior Manager	315	20.41
	Chief Executive	82	5.31
Company Tenure	Less than a year	28	1.81
	Between 1 and 3 Years	288	18.66
	Between 3 and 5 Years	552	35.77
	Between 5 and 10 Years	435	28.19
	More than 10 years	240	15.55

noting that all item loadings were statistically significant with the lowest standardised loading being 0.667 ($p < 0.01$). Moreover, convergent validity was established as all constructs had Average Variance Extracted (AVE) values surpassing the value of 0.5. Discriminant validity was confirmed by ensuring that each constructs AVE exceeded its Maximum Shared Variance (MSV) and by verifying distinctiveness among factors, through correlation comparisons. The Correlation Matrix analysis further supported these findings. The model's fit was evaluated using several indices: root mean square error of approximation (RMSEA) at.026, comparative fit index (CFI) at.952, Tucker- is index (TLI) at.949, incremental fit index (IFI) at.953, Chi-square at 3855.368, and Degrees of freedom at 2713. The significant systematic variance among the individual indicators, attributed to the underlying latent construct, lends empirical validation to our scale. Furthermore, common method variance analysis (CMV) was used to validate that the observed correlations between the independent and dependent variables of the study are not biased by the single-method survey data collection. A standard procedure involving Harman's one-factor analysis and common latent factor (CLF) was performed. The results revealed no predominant factor and a level of variance well below the 50 % threshold, while the CLF displayed significant statistical differences and a low level of shared variance, suggesting no concerns around common method bias affecting the validity of our findings.

5.2.4. Structural equation modelling (SEM)

The last stage employed SEM to test the conceptual model (Fig. 1), with maximum likelihood estimation of our proposed hypotheses. The model's fitness was evaluated on several indices. A chi-square value of 6283.782, though it is a traditional measure for significance, is subjective to sample size which necessitates that we use alternative indices for more robust assessment too. Comparative Fit Index (CFI =.957), Tucker-Lewis Index (TLI =.956) and Incremental Fit Index (IFI =.958) all have values above the generally accepted threshold of.950 indicating a good fit between the hypothesised model and the observed data. These numbers are not so much influenced by sample size, and thus point out how much variance or covariance in the data has been explained by the model. Moreover, Root Mean Square Error of Approximation (RMSEA=.018<0.05) shows a good fit of the model to observed data. Finally, the results from the estimated structural pathways revealed significant connections and all 4 hypotheses were supported, as follows: Employee preparedness and employee AI acceptance (Hypothesis 1; γ =.414, $t = 2.013$, $p < 0.05$), employee AI acceptance and AI adoption (Hypothesis 2; $\gamma = 1.809$, $t = 2.216$, $p < 0.05$), AI adoption and SDG adoption (Hypothesis 3; γ =.049, $t = 3.180$, $p < 0.01$), and SDG adoption and Non-financial performance (Hypothesis 4; $\gamma = 1.928$, $t = 3.413$, $p < 0.01$).

6. Discussion

The current research sheds light on important knowledge gaps in hospitality studies. By developing and validating a novel scale to explore how interrelationships between AI, SDG adoption and NFP, in the hospitality context. Prior studies (e.g., Kline et al., 2023) have provided some valuable insight into the importance of AI and SDGs in the hospitality context, but no study has previously provided empirical insight into SDG adoption in the independent hotel context. As such, considering the importance of SDGs and AI (e.g., MacInnes et al., 2022) in hospitality, such lack of conceptualisation and operationalisation is surprising.

Therefore, this study enhances understanding among scholars, hotel managers, and policymakers about the role of AI in achieving Sustainable Development Goals (SDGs). It addresses two main questions: (i) Do factors including employee readiness and acceptance of AI (akin to commitment to learn and behavioural intentions) influence AI adoption and its contribution to SDG achievement? (ii) Does successful AI adoption in hotels influence NFP?

The results indicated that all of the hypotheses tested were supported. The results first demonstrate that AI adoption is driven by employees' technological readiness and employees' acceptance of AI technology (H1 and H2), therefore highlighting the importance of encouraging employees to understand and perceive the benefits that a new disruptive technology can bring. Further, the results confirm a significant contribution of AI adoption to employees' adoption of SDG principles (H3), which in turn positively influences the organisation's commitment to SDGs. Finally, the results (from H4) also show a strong statistical significance indicating that SDG adoption positively relates to the three dimensions of NFP measured herein, showing that adopting sustainable practices within an organisation's operations contribute to the employees' level of optimism, satisfaction and even engagement with AI (H4).

Focusing on the independent hospitality context in the US, these results offer a perspective of how AI and SDG adoption could be implemented, supporting introduction of these practices within the organisations' operations. These results demonstrate that in the hospitality context, and especially that of independent hotels, what employees would truly benefit from and could influence their behavioural intentions towards AI adoption is the perceived improved self-efficacy, such as from using AI to automate tasks and enable better decision-making. These results are consistent with Buhalis et al. (2019) and Tschang and Almirall (2021) who offer insights into how organisations and some hotels are adopting AI in their environments and specifically the importance of proper training into AI systems, and how to leverage data-driven insights into enhanced performance. In addition, this research suggests that to facilitate AI acceptance, hotels should provide clear communication to employees to mitigate any concerns about job security and invest in targeted training programs.

Moreover, regarding AI's role in SDG adoption within the independent hotel industry, the results suggest how novel technology could be used to allow organisations to keep up with evolving stakeholder demands and preferences, increasingly aligning with the SDGs principles, as Barthel et al. (2015) discuss. Furthermore, the strong correlation between AI adoption and the SDGs suggests that independent hotels' adoption of AI can contribute to broader sustainability aims, such as efficient use of resources, waste reduction, and promotion of environmental protection by (for example) increasing energy efficiency and lowering operational costs. The development of AI-driven policies in order to increase overall efficiency, which directly affects economic growth, is one of the most important reasons to adopt it, as discussed by Manika et al. (2013).

Finally, a major finding of this research is that the adoption of the SDGs, driven by increased AI adoption, can positively affect the NFP of organisations. In relation to independent hotels, increased SDG adoption may increase employee satisfaction and optimism due to perceived purpose and satisfaction in contributing to broader social and environmental goals, consistent with discussions by Rydzik and Kissoon (2021) and Guix and Font (2020). Additionally, this finding evidenced also that the strong correlation observed between AI adoption and improvements in NFP metrics such as employee optimism, satisfaction and engagement towards AI could enable for example employees' quality of work or productivity (Items NFP1 and NFP4).

The present study clearly shows the significant interrelationships around increased employee engagement with both SDG practices and AI. For example, seeing the impacts in practice may lead employees to better embrace the use of AI in more of their activities, by contributing with ideas for its better utilisation, and adapting more quickly to changes in their work environment. However, apart from making employees more satisfied with their roles and the prospects AI could bring to their future, organisations must also understand that such benefits require an investment of time and resources, which need to be accounted for when deciding to implement AI into their operations.

7. Conclusion

7.1. Theoretical contribution

The importance of AI and SDG adoption in hospitality has been raised by the pre-existing literature. However, it did not provide any empirical and systematic insight into how such adoption should be conceptualised or measured. This research uniquely conceptualises and operationalises AI and SDG adoption in hospitality with a new composite scale which adapted items from previous work, focusing on independent hotels, thus filling the gap left by prior research. Thus, the scale developed in this study contributes significantly to the literature by demonstrating how relevant aspects of SDG adoption (i.e. from organisations' commitment to clean energy, zero hunger, and promoting social equality) can be influenced and catalysed by the use of and adoption of AI (see Appendix B for more details on the constructs).

A novel mechanism through which AI and SDG adoption antecedents influence diverse outcomes was also tested in this research. Similar to previous studies (e.g., Oyman et al., 2022) our research showed that employee readiness positively impacts on employee acceptance of AI. However, our study furthered this understanding by also showing that employee acceptance of AI can impact on AI adoption. Furthermore, building on previous studies (e.g., Yoon, 2018) and the TAM theoretical framework (Davis, 1989) our results showed that adoption of, and intention toward, new technologies can be enhanced through employee understanding of their usefulness and perceived ease of use, therefore enriching existing theories, by providing an AI perspective.

Digital technologies provide opportunities for societies and businesses to address great global challenges, such as poverty, pandemics and climate change. Our results showed that AI adoption can significantly impact SDG adoption, by offering innovative solutions to address complex challenges, contributing to sustainable development across various sectors. Lastly, our findings showed that SDG adoption can positively influence NFP through enhancing employee optimism, satisfaction and AI engagement (Lu et al., 2023), highlighting the link to employee wellbeing and satisfaction. Taken together, these findings extend the boundaries of existing literature by empirically showing the antecedents and consequences of AI and SDG adoption, each operating through distinct mechanisms, as shown by the significant findings and relationships of our conceptual model.

7.2. Managerial implications

Our study highlighted important practical implication for hospitality managers who are considering adopting AI and SDGs in their business activities. Hospitality managers can use our developed measures to capture AI adoption and employ our novel, validated scale to understand how employees perceive firms' contribution to SDGs. As such, by using the scale, firms can glean a comprehensive insight into overall impact of SDG adoption and related activities into their performance, for example through managers assessing employees' perceptions of the hotel's SDG efforts and their impact through regular surveys or feedback sessions. Our results also highlighted the importance of AI adoption in facilitating SDG adoption, calling upon the development of strategies and policymaking for AI implementation that allow managers to familiarise staff with AI benefits and ease of use of these platforms first, then gradually introducing more complex systems to achieve the organisation's sustainability goals. In the hospitality industry, AI can bring about significant innovation in the development of sustainable infrastructure across most hotel departments and SDG aspects, by enhancing guest experiences, optimising operations, and improving overall efficiency (Chi et al., 2022). At an operational level, AI algorithms can manage energy and water consumption by adjusting lighting, temperature, and other systems based on occupancy patterns (Bi et al., 2022).

Likewise, AI adoption can help manage inventory and predict consumption patterns in dining and housekeeping operations (Elkhwesky et al., 2022), or help employees to effectively schedule housekeeping services based on guest preferences and occupancy, thus reducing unnecessary wastage (Kim et al., 2012). AI can also contribute to greater gender equality and reducing inequality by promoting equal opportunities between employees and creating a more inclusive and diverse work environment (Xiang et al., 2021), e.g. by helping with fair scheduling, unbiased recruitment, or training programs. At a strategic level as well, AI-driven algorithms can be used in hotels to analyse environmental data to aid in climate monitoring patterns and suggest adaptation strategies and AI-driven feedback systems can allow guests to share suggestions and concerns regarding sustainable practices, fostering a sense of collaboration between guests and the hotel management and uncovering ways for further improvement. Even at an employees' individual level, an AI algorithm can suggest career paths and development opportunities to employees based on their skills and aspirations, ensuring that individuals of all genders have equal access to growth prospects, therefore demonstrating its usefulness across all sectors and level of deployment within an organisation.

7.3. Limitations and future research

The conclusions drawn from our study should be considered within the context of certain limitations arising from the choices made in the research design. This research was carried out in a developed market where there is a well-established rise in companies' desire for adoption of new technologies and SDGs. Nonetheless, it remains uncertain whether hotels in emerging markets would demonstrate a similar upsurge in the adoption of AI and SDGs. Thus, future studies are advised to take a cross-cultural analysis approach and investigate how different country contexts can impact on our findings. Future studies can compare their findings across different countries to see if their findings differ across different cultures and environments. Furthermore, upcoming research endeavours have the potential to explore the influence of various other elements on AI and SDG adoption not extensively discussed in this paper (for example other employee characteristics impacting their AI adoption, such as age and level of education), and perhaps more relevant in other tourism destinations. Moreover, additional investigation is needed into the way employees from different hotel categories perceive AI and SDG adoption and whether these perceptions differ under diverse circumstances (e.g. from luxury to boutique, to all-inclusive to budget). Also, to provide a more accurate account of employee behaviour, longitudinal studies could track changes in employee readiness for AI an impacts to SDG adoption and NFP over longer periods of time, potentially creating specific case studies which can be further leveraged in the creation of a technology acceptance model in hospitality and tourism. These research areas would provide a deeper understanding of the intersections between AI, SDGs, and NFP in the hospitality industry and offer further practical insights for hotel managers and policymakers, built on the foundations provided by this paper.

During the preparation of this work the author used ChatGPT and Microsoft Copilot to more efficiently organise and format references, check specific words/sentences translation and ensure the paper does not have any grammatical or spelling errors, as the author is not a native English speaker. After using this tool/service, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

CRediT authorship contribution statement

Maria JesusJerez-Jerez: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration.

Declaration of Competing Interest

The author has no conflict of interest in this work.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.ijhm.2024.103987.

References

Anderson, M., Bolton, J., 2015. Integration of sensors to improve customer experience: Implementing device integration for the retail sector. In 2015 (October). IEEE 12th International Conference on e-Business Engineering. IEEE, pp. 382–386 (October).

Barthel, R., Hudson-Smith, A., de Jode, M., 2015. London global university, UCL. Future Retail Environ. Internet things (IoT).

Bhusan, R.M.B., 2024. The spectre of generative AI over advertising, marketing, and branding. J. Future Media Stud. 1 (1), 1–10 (DOI: pending).

Bi, J.W., Li, C., Xu, H., Li, H., 2022. Forecasting daily tourism demand for tourist attractions with big data: an ensemble deep learning method. J. Travel Res. 61 (8), 1719–1737. https://doi.org/10.17632/xwj98nb39r.1.

Brakus, J.J., Schmitt, B.H., Zarantonell, L., 2009. Brand experience: what is it? How is it measured? Does it affect loyalty? J. Mark. 73 (3), 52–68.

Buhalis, D., Harwood, T., Bogicevic, V., Viglia, G., Beldona, S., Hofacker, C., 2019. Technological disruptions in services: lessons from tourism and hospitality. J. Serv. Manag. 30 (4), 484–506.

Buhalis, D., Leung, R., 2018. Smart hospitality—interconnectivity and interoperability towards an ecosystem. Int. J. Hosp. Manag. 71, 41–50.

Buhalis, D., O'Connor, P., Leung, R., 2023. Smart hospitality: from smart cities and smart tourism towards agile business ecosystems in networked destinations. Int. J. Contemp. Hosp. Manag. 35 (1), 369–393.

Cakir, G., Bezbradica, M., Helfert, M., 2019. The Shift from financial to non-financial measures during transition into digital retail–a systematic literature review. June 26–28, 2019, Proceedings, Part I. In: Business Information Systems: 22nd International Conference, BIS 2019, 22. Springer International Publishing, Seville, Spain, pp. 189–200. June 26–28, 2019, Proceedings, Part I.

Chan, E.K., Wan, L.C., 2023. How eco-certificate/effort influences hotel preference. Ann. Tour. Res. 101, 103616.

Chang, V., Ramachandran, M., Yao, Y., Kuo, Y.H., Li, C.S., 2016. A resiliency framework for an enterprise cloud. Int. J. Inf. Manag. 36 (1), 155–166.

Chatterjee, S., Rana, N.P., Dwivedi, Y.K., Baabdullah, A.M., 2021. Understanding AI adoption in manufacturing and production firms using an integrated TAM-TOE model. Technol. Forecast. Soc. Change 170, 120880.

Chi, O.H., Gursoy, D., Chi, C.G., 2022. Tourists' attitudes toward the use of artificially intelligent (AI) devices in tourism service delivery: moderating role of service value seeking. J. Travel Res. 61 (1), 170–185.

Cuervo-Cazurra, A., Dieleman, M., Hirsch, P., Rodrigues, S.B., Zyglidopoulos, S., 2021. Multinationals' misbehavior. J. World Bus. 56 (5), 101244.

Davis, F.D. (1989). Technology acceptance model: TAM. In Al-Suqri, MN, Al-Aufi, AS: Information Seeking Behavior and Technology Adoption, 205-219.

Denscombe, M., 2007. The Good Research Guide: For Small-Scale Social Research Projects, 2nd ed. Open University Press, Maidenhead, UK.

Di Vaio, A., Varriale, L., Di Gregorio, A., Adomako, S., 2022. Corporate social performance and non-financial reporting in the cruise industry: paving the way towards UN Agenda 2030. Corp. Soc. Responsib. Environ. Manag. 29 (6), 1931–1953.

Du, S., El Akremi, A., Jia, M., 2022. Quantitative research on corporate social responsibility: a quest for relevance and rigor in a quickly evolving, turbulent world. J. Bus. Ethics 1–15.

Elkhwesky, Z., El Manzani, Y., Elbayoumi Salem, I., 2022. Driving hospitality to foster sustainable innovation: a systematic review of COVID-19-related studies and practical implications in the digital era. Tourism and Hospitality. Research, 14673584221126792.

Foroudi, P., Gupta, S., Sivarajah, U., Broderick, A., 2018. Investigating the effects of smart technology on customer dynamics and customer experience. Comput. Hum. Behav. 80, 271–282.

Foroudi, P., Marvi, R., Cuomo, M.T., Bagozzi, R., Dennis, C., Jannelli, R., 2023. Consumer perceptions of sustainable development goals: conceptualization, measurement and contingent effects. Br. J. Manag. 34 (3), 1157–1183.

Gonzalez-Arcos, C., Joubert, A.M., Scaraboto, D., Guesalaga, R., Sandberg, J., 2021. "How do I carry all this now?" Understanding consumer resistance to sustainability interventions. J. Mark. 85 (3), 44–61.

Guix, M., Font, X., 2020. The materiality balanced scorecard: a framework for stakeholder-led integration of sustainable hospitality management and reporting. Int. J. Hosp. Manag. 91, 102634.

Haidt, J., 2007. The new synthesis in moral psychology. Science 316 (5827), 998–1002.

Hair, J.F., Sarstedt, M., Ringle, C.M., Gudergan, S.P., 2018. Advanced Issues in Partial Least Squares Structural Equations Modeling (PLS-SEM). Sage, Thousand Oaks, CA.

Hotel Tech Report. (2024). Hospitality statistics. Retrieved May 10, 2024, from https://hoteltechreport.com/news/hospitality-statistics.

Hwang, Y., Shi, X., Wang, X., 2021. Hospitality employees' emotions in the workplace: a systematic review of recent literature. Int. J. Contemp. Hosp. Manag. 33 (10), 3752–3796.

Islam, T., Ahmad, S., Kaleem, A., Mahmood, K., 2021. Abusive supervision and knowledge sharing: moderating roles of Islamic work ethic and learning goal orientation. Manag. Decis. 59 (2), 205–222.

Jang, H.W., Yoo, J.J.E., Cho, M., 2024. Resistance to blockchain adoption in the foodservice industry: moderating roles of public pressures and climate change awareness. Int. J. Contemp. Hosp. Manag. 36 (5), 1467–1489.

Kamal, S.A., Shafiq, M., Kakria, P., 2020. Investigating acceptance of telemedicine services through an extended technology acceptance model (TAM). Technol. Soc. 60, 101212.

Kim, K.A., Byon, K.K., Baek, W., Williams, A.S., 2019. Examining structural relationships among sport service environments, excitement, consumer-to-consumer interaction, and consumer citizenship behaviors. Int. J. Hosp. Manag. 82, 318–325.

Kim, Y.J., Palakurthi, R., Hancer, M., 2012. The environmentally friendly programs in hotels and customers' intention to stay: an online survey approach. Int. J. Hosp. Tour. Adm. 13 (3), 195–214.

Kline, C.S., Hoarau-Heemstra, H., Cavaliere, C.T., 2023. Wildlife equity theory for multispecies tourism justice. J. Travel Res. 62 (6), 1167–1180.

Legrand, W., Matthew-Bolofinde, A., 2022. ESGs, SDGs, and hospitality: challenges and opportunities in activating sustainability. Bus. 21st Century.: A Sustain. Approach 25–39.

Li, M., Yin, D., Qiu, H., Bai, B., 2021. A systematic review of AI technology-based service encounters: Implications for hospitality and tourism operations. Int. J. Hosp. Manag. 95, 102930.

Lim, J., Vaughan, Y., Jang, J., 2023. Do employees' perceptions of diversity management enhance firm's financial performance: the moderating role of board members' diversity level. Int. J. Contemp. Hosp. Manag. 35 (11), 3990–4009.

Lu, Y., Zhang, M.M., Yang, M.M., Wang, Y., 2023. Sustainable human resource management practices, employee resilience, and employee outcomes: toward common good values. Hum. Resour. Manag. 62 (3), 331–353.

MacInnes, S., Grün, B., Dolnicar, S., 2022. Habit drives sustainable tourist behaviour. Ann. Tour. Res. 92, 103329.

Manyika, J., Chui, M., Bughin, J., Dobbs, R., Bisson, P., Marrs, A., 2013. Disruptive technologies: Advances that will transform life, business, and the global economy, 180. McKinsey Global Institute, San Francisco, CA, pp. 17–21.

Molina-Collado, A., Santos-Vijande, M.L., Gómez-Rico, M., Madera, J.M., 2022. Sustainability in hospitality and tourism: a review of key research topics from 1994 to 2020. Int. J. Contemp. Hosp. Manag. 34 (8), 3029–3064.

Montiel, I., Cuervo-Cazurra, A., Park, J., Antolín-López, R., Husted, B.W., 2021. Implementing the United Nations' sustainable development goals in international business. J. Int. Bus. Stud. 52 (5), 999–1030.

Murphy, R., 2000. Introduction to AI robotics. MIT Press, Cambridge, MA.

Nam, K., Dutt, C.S., Chathoth, P., Daghfous, A., Khan, M.S., 2020. The adoption of artificial intelligence and robotics in the hotel industry: prospects and challenges. J. Revenue Pricing Manag. 19 (6), 469–483.

Nazarian, A., Atkinson, P., Foroudi, P., Dennis, K., 2019. Finding the right management approach in independent hotels. Int. J. Contemp. Hosp. Manag. 31 (7), 2862–2883.

Nilsson, M., Chisholm, E., Griggs, D., Howden-Chapman, P., McCollum, D., Messerli, P., Stafford-Smith, M., 2018. Mapping interactions between the sustainable development goals: lessons learned and ways forward. Sustain. Sci. 13, 1489–1503.

Oriade, A., Osinaike, A., Aduhene, K., Wang, Y., 2021. Sustainability awareness, management practices and organisational culture in hotels: evidence from developing countries. Int. J. Hosp. Manag. 92, 102699.

Oyman, M., Bal, D., Ozer, S., 2022. Extending the technology acceptance model to explain how perceived augmented reality affects consumers' perceptions. Comput. Hum. Behav. 128, 107127.

Pantano, E., Timmermans, H., 2014. What is smart for retailing? Procedia Environ. Sci. 22, 101–107.

Rather, R.A., Hollebeek, L.D., Loureiro, S.M.C., Khan, I., Hasan, R., 2023. Exploring tourists' virtual reality-based brand engagement: a uses-and-gratifications perspective. J. Travel Res. 63 (3), 606–624.

Ribeiro, M.A., Gursoy, D., Chi, O.H., 2022. Customer acceptance of autonomous vehicles in travel and tourism. J. Travel Res. 61 (3), 620–636.

de Ruyter, K., Keeling, D.I., Plangger, K., Montecchi, M., Scott, M.L., Dahl, D.W., 2022. Reimagining marketing strategy: driving the debate on grand challenges. J. Acad. Mark. Sci. 50 (1), 13–21.

Rydzik, A., Kissoon, C.S., 2021. Decent work and tourism workers in the age of intelligent automation and digital surveillance. J. Sustain. Tour. 30 (12), 2860–2877.

Salameh, A.A., Aman-Ullah, A., Mehmood, W., Abdul-Majid, A.H.B., 2023. Does employer branding facilitate the retention of healthcare employees? A mediation moderation study through organisational identification, psychological involvement, and employee loyalty. Int. J. Hosp. Manag. 112, 103414.

Salehi, M., Filimonau, V., Ghaderi, Z., Hamzehzadeh, J., 2021. Energy conservation in large-sized hotels: Insights from a developing country. Int. J. Hosp. Manag. 99, 103061.

Skandali, D., Magoutas, A., Tsourvakas, G., 2024. Consumer behaviour on AI applications for services: measuring the impact of value-based adoption model on luxurious AI resorts' applications. Rev. Mark. Sci. https://doi.org/10.1515/roms-2023-0099.

Smerecnik, K.R., Andersen, P.A., 2011. The diffusion of environmental sustainability innovations in North American hotels and ski resorts. J. Sustain. Tour. 19 (2), 171–196.

Stylos, N., Fotiadis, A.K., Shin, D.D., Huan, T.C.T., 2021. Beyond smart systems adoption: Enabling diffusion and assimilation of smartness in hospitality. Int. J. Hosp. Manag. 98, 103042.

Sun, J.J., Bellezza, S., Paharia, N., 2021. Buy less, buy luxury: understanding and overcoming product durability neglect for sustainable consumption. J. Mark. 85 (3), 28–43.
Sun, Z., Wu, L.Z., Ye, Y., Kwan, H.K., 2023. The impact of exploitative leadership on hospitality employees' proactive customer service performance: a self-determination perspective. Int. J. Contemp. Hosp. Manag. 35 (1), 46–63.
Tschang, F.T., Almirall, E., 2021. Artificial intelligence as augmenting automation: Implications for employment. Acad. Manag. Perspect. 35 (4), 642–659.
Venkatesh, V., Bala, H., 2008. Technology acceptance model 3 and a research agenda on interventions. Decis. Sci. 39 (2), 273–315.
de Visser-Amundson, A., 2022. A multi-stakeholder partnership to fight food waste in the hospitality industry: a contribution to the United Nations Sustainable Development Goals 12 and 17. J. Sustain. Tour. 30 (10), 2448–2475.
Wang, W., Chen, L., Xiong, M., Wang, Y., 2023. Accelerating AI adoption with responsible AI signals and employee engagement mechanisms in health care. Inf. Syst. Front. 25 (6), 2239–2256.
Wang, Y., So, K.K.F., Sparks, B.A., 2017. Technology readiness and customer satisfaction with travel technologies: a cross-country investigation. J. Travel Res. 56 (5), 563–577.
Xiang, L., Wang, P., Yang, X., Hu, A., Su, H., 2021. Fault detection of wind turbine based on SCADA data analysis using CNN and LSTM with attention mechanism. Measurement 175, 109094.
Yang, Z., Sun, J., Zhang, Y., Wang, Y., 2015. Understanding SaaS adoption from the perspective of organizational users: a tripod readiness model. Comput. Hum. Behav. 45, 254–264.
Yoon, C., 2018. Extending the TAM for Green IT: a normative perspective. Comput. Hum. Behav. 83, 129–139.
Zalama, E., García-Bermejo, J.G., Marcos, S., Domínguez, S., Feliz, R., Pinillos, R., López, J., 2014. Sacarino: Implementing a service robot in a hotel environment (November). In: ROBOT2013: First Iberian Robotics Conference: Advances in Robotics, 2. Springer International Publishing, pp. 3–14 (November).

Hospitality & Tourism Management expert with 30+ years of experience. Senior Lecturer In International Business at the University of Bedfordshire. PhD in Business Management. Senior Fellow of the Higher Education Academy. My current research interest is focused on disruptive innovation in the hospitality field.